PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Efficacious of dexmedetomidine in children undergoing cleft lip and palate repair: a systematic review and meta-analysis
AUTHORS	Liu, Dengfeng; Pan, Li; Gao, Yin; Liu, Jiefan; Li, Feng; Li, Xiangwei; Quan, Jiale; HUANG, CONGCONG; Lian, Chunwei

VERSION 1 – REVIEW

REVIEWER	Ghimire, Laxmi Tribhuvan University Teaching Hospital, Department of Internal Medicine
REVIEW RETURNED	08-Dec-2020

GENERAL COMMENTS	Title: I think it could be more clear, like what was dexmedetomidine efficacious in? Also, I assume it is systematic review and meta-analysis, not just system review. Methods: differences in the doses(some are multiple fold of others), timing of administration(some throughout the surgery, while others at the very end of surgery), and with different controls(saline, and other anesthesia medications) makes this meta-analysis very challenging to draw conclusions from. Also, the different routes of administration(intranasal, perineural, intravenous), which could lead to differences in pharmacokinetics and varied response. The included studies have children as young as 3 months to as old as 12 years. Were dexmedetomidine level drawn for any of the studies? This is also important to some extent as receptors and dexmedetomidine metabolizing enzymes systems might not be as mature in very young children(esp infants) compared to older children. This could affect the efficacy and ultimately the outcomes . Dexmedetomidine has been used as an adjunct drug along with other anesthetic agents in the studies used for this review/meta-analysis. I don't get this sense while reading this manuscript except for being mentioned in the abstract. I would suggest the authors clarify whether dexmedetomidine was used alone or as an adjunct to other anesthetic. In table 1, it might be helpful if the authors include what other anesthetic agents were used in each study in addition to dexmedetomidine. Also mention, if Dexmedetomidine was used as the sole anesthetic agent.

	Choosing between fixed effect model vs random effect model: I think default should be random effect model unless the studies were performed in the same setting/hospital, using the same personnel, and similar baseline characteristics. Otherwise, random effect model analysis is more appropriate.

REVIEWER	Turcu-Stiolica, Adina University of Medicine and Pharmacy of Craiova, Department of Biostatistics
REVIEW RETURNED	21-Dec-2020

GENERAL COMMENTS	Please do the figure for Risk of bias with all the information from page 33 to 57. Please do the subgroup analysis after administered way: intravenous, intranasal and perineural administration. You presented the analysis ”Perioperative dexmedetomidine versus control groups for emergence agitation (EA)” in figure 2. But you also presented in fig. 2 the analysis of dexmedetomidine versus fentanyl and ketamine. And also you have Figure 1, 2, 3 and 4 for funnel plots that are not discussed. Please discuss the funnel plots! You present some results with p-value and I2 and some results only with p-value. It is possible to interpret the p-value for overall effect or for heterogeneity. I think it's better to present the outcomes in a unique way, because you don't have all the forest plots to see what p-value you present.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

-Title: I think it could be more clear, like what was dexmedetomidine efficacious in? Also, I assume it is systematic review and meta-analysis, not just system review.
Response: We have revised the title as “Efficacious of dexmedetomidine in children during cleft lip and palate repair: a systematic review and meta-analysis”. Thanks.

-Methods: differences in the doses(some are multiple fold of others), timing of administration(some throughout the surgery, while others at the very end of surgery), and with different controls(saline, and other anesthesia medications) makes this meta-analysis very challenging to draw conclusions from.
Response: Yes, it is one of our limitations. The conclusion of a systematic review was based on original study. More high-quality studies would be needed in the future. But our systematic review identified the benefits of dexmedetomidine in children during CLP repair. Thanks.

-Also, the different routes of administration(intranasal, perineural, intravenous), which could lead to differences in pharmacokinetics and varied response.
Response: Yes, it is right. We have added the subgroup analysis. Thanks.

-The included studies have children as young as 3 months to as old as 12 years. Were dexmedetomidine level drawn for any of the studies? This is also important to some extent as receptors and dexmedetomidine metabolizing enzymes systems might not be as mature in very young children(esp infants) compared to older children. This could affect the efficacy and ultimately the outcomes .
Response: The dexmedetomidine level was not drawn in any of our included studies. Dexmedetomidine appeared to exhibit a level of efficacy similar to that seen in adults and to be fairly well tolerated in children aged older than 1 month (DOI: 10.1007/s40262-017-0507-7).

-Dexmedetomidine has been used as an adjunct drug along with other anesthetic agents in the studies used for this review/meta-analysis. I don't get this sense while reading this manuscript except for being mentioned in the abstract. I would suggest the authors clarify whether dexmedetomidine was used alone or as an adjunct to other anesthetic. In table 1, it might be helpful if the authors include what other anesthetic agents were used in each study in addition to dexmedetomidine. Also mention, if Dexmedetomidine was used as the sole anesthetic agent.
Response: We have added other anesthetic agents which were used in each study in addition to dexmedetomidine in table 1. Thanks.

-Choosing between fixed effect model vs random effect model: I think default should be random effect model unless the studies were performed in the same setting/hospital, using the same personnel, and similar baseline characteristics. Otherwise, random effect model analysis is more appropriate.
Response: Thanks for your advice. we have checked it and revised with the random effect model analysis in partial results of outcomes (Respiratory adverse events, Cardiovascular adverse events, PONV).
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Reviewer: 2
-Please do the figure for Risk of bias with all the information from page 33 to 57.
Response: We have added the figure for the risk of bias (as fig 2). Thanks.

-Please do the subgroup analysis after administered way: intravenous, intranasal and perineural administration.
Response: We have added the subgroup analysis. Thanks.

-You presented the analysis ”Perioperative dexmedetomidine versus control groups for emergence agitation (EA)” in figure2. But you also presented in fig. 2 the analysis of dexmedetomidine versus fentanyl and ketamine.
Response: In our article, control groups included saline control and other drug control. Thanks.

-And also you have Figure 1,2, 3 and 4 for funnel plots that are not discussed. Please discuss the funnel plots!
Response: Because all the numbers of the included studies of our comparisons were less than ten, it was difficult to evaluate its symmetry of our funnel plots. Much more studies would be needed in the future. Thanks.

-You present some results with p-value and I2 and some results only with p-value. It is possible to interpret the p-value for overall effect or for heterogeneity. I think it's better to present the outcomes in a unique way, because you don't have all the forest plots to see what p-value you present.

Response: We have checked all the results. The results only with p-value because of their p-values were greater than 0.05. We have revised them and presented all the outcomes in the same way except the ones whose heterogeneity were not applicable. Thanks.

VERSION 2 – REVIEW

REVIEWER	Ghimire, Laxmi Tribhuvan University Teaching Hospital, Department of Internal Medicine
REVIEW RETURNED	03-Mar-2021

GENERAL COMMENTS	Authors have tried to address all the concerns and suggestions that I made to improve on the manuscript.

VERSION 2 – AUTHOR RESPONSE

Reviewer 1

-Authors have tried to address all the concerns and suggestions that I made to improve on the manuscript.
Response: We have tried to address all the questions the reviewer asked. thanks.